Divisibility as Arithmetic in Quotient Rings

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The most productive step in many divisibility problems is to stop treating the divisor as a large integer to be compared with an even larger expression. Once the divisor m is fixed, calculations take place among residue classes in $\mathbb{Z}/m\mathbb{Z}$. The familiar operations of greatest common divisors, Bezout coefficients, and the Chinese remainder theorem describe the structure of that ring.[1]

§1.1 A necessary condition can collapse the search

Suppose a prime has the form

$$p = x(x+1) - 4, \qquad x \in \mathbb{Z}_{>0}.$$

The product of consecutive integers is even, so p is even. The only even positive prime is 2. Therefore $x(x+1) = 6$ and $x = 2$, which indeed gives $p = 2$. The short argument has two parts: parity narrows the candidates, and substitution verifies the surviving candidate.

A similar reduction works even when the divisor depends on the unknown integer. Consider

$$x^2 + 1 \mid x^3 - x, \qquad x \in \mathbb{Z}_{>0}.$$

Since $x^2 \equiv -1 \pmod{x^2+1}$, the condition is equivalent to

$$x^2 + 1 \mid 2x.$$

For $x > 1$, $x^2 + 1 > 2x > 0$, making divisibility impossible. At $x = 1$, the dividend vanishes, so $x = 1$ is the unique positive solution.

§1.2 Ideals explain why remainders preserve information

For nonzero $m \in \mathbb{Z}$, the congruence $a \equiv b \pmod{m}$ means $a - b \in m\mathbb{Z}$. In the quotient ring $\mathbb{Z}/m\mathbb{Z}$, that is simply equality of the classes $[a] = [b]$. Polynomial or exponential expressions respect this equality, so we can simplify them before asking whether the original expression is divisible by m.

The ideal $(a) = a\mathbb{Z}$ contains the multiples of a, and

$$a \mid b \quad \Longleftrightarrow \quad (b) \subseteq (a).$$

The reversal of inclusion is useful. Bezout's identity implies

$$(a) + (b) = (\gcd(a,b)), \qquad (a) \cap (b) = (\operatorname{lcm}(a,b))$$

for nonzero integers, using positive gcd and lcm. Thus gcd and lcm describe two universal ways to combine divisibility conditions.

A unit in $\mathbb{Z}/m\mathbb{Z}$ is a class $[a]$ possessing a multiplicative inverse. Such an inverse exists exactly when $\gcd(a,m) = 1$: the Bezout equation $au + mv = 1$ becomes $[a][u] = [1]$. By contrast, modulo 6, the nonzero classes [2] and [3] multiply to zero. This zero-divisor phenomenon explains why cancellation by an arbitrary nonzero class is invalid.

[1] See the introductory number theory and congruence lectures of MIT 18.781: https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/.

§1.3 A polynomial remainder hidden in an integer problem

For integer x, consider

$$x^2 + x + 1 \mid x^4 - 2.$$

The identity

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

shows that $x^3 \equiv 1 \pmod{x^2 + x + 1}$, whence

$$x^4 - 2 \equiv x - 2 \pmod{x^2 + x + 1}.$$

For positive x, the divisor exceeds $|x - 2|$ except when $x = 2$, and the small remaining value $x = 1$ also fails. At $x = 2$, $x^4 - 2 = 14$ and $x^2 + x + 1 = 7$, so $x = 2$ is the unique positive solution.

There is a parallel computation in the polynomial quotient $k[t]/(t^2 + t + 1)$, where the symbol t satisfies $t^3 = 1$. That statement is about a class of a polynomial, independent of any integer substituted for t. The integer problem evaluates the same polynomial identity at a particular x, then interprets it as a congruence. Keeping these two uses of a symbol separate prevents a common logical slip.

§1.4 The Chinese remainder theorem builds solutions

Suppose m, n are coprime. Choose $u, v \in \mathbb{Z}$ satisfying $um + vn = 1$. Then the classes

$$e_m = vn, \qquad e_n = um$$

satisfy

$$e_m \equiv 1 \pmod{m}, \quad e_m \equiv 0 \pmod{n}, \qquad e_n \equiv 0 \pmod{m}, \quad e_n \equiv 1 \pmod{n}.$$

Therefore the simultaneous system

$$x \equiv a \pmod{m}, \qquad x \equiv b \pmod{n}$$

has solution

$$\boxed{x \equiv ae_m + be_n \pmod{mn}.}$$

Any two solutions differ by a multiple of both m and n, hence by a multiple of mn; the solution is unique modulo mn.

For example, solve $x \equiv 2 \pmod 3$ and $x \equiv 4 \pmod 5$. The selectors $10 \equiv 1 \pmod 3$, $0 \pmod 5$ and $6 \equiv 0 \pmod 3$, $1 \pmod 5$ give

$$x \equiv 2 \cdot 10 + 4 \cdot 6 = 44 \equiv 14 \pmod{15}.$$

These selectors are idempotents: modulo 15, $10^2 \equiv 10$, $6^2 \equiv 6$, $10 \cdot 6 \equiv 0$, and $10 + 6 \equiv 1$. Their algebraic behavior is exactly that of coordinate projections in the product ring

$$\mathbb{Z}/15\mathbb{Z} \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}.$$

The general theorem for pairwise coprime moduli follows by the same Bezout construction. It is an explicit decomposition, rather than a mysterious claim that local solutions can be assembled globally.

§1.5 Odd powers and the arithmetic of a reflection

A large example is

$$S = 1^{1983} + 2^{1983} + \cdots + 1983^{1983}.$$

The claim is that S is divisible by the triangular number

$$T = \frac{1983 \cdot 1984}{2} = 1983 \cdot 992.$$

The parity of the number of terms matters. Since 1983 is odd, pairing j with $1983 - j$ for $1 \leq j \leq 1982$ gives $1983 \mid S$. Pairing instead j with $1984 - j$ gives

$$j^{1983} + (1984 - j)^{1983} \equiv 0 \pmod{992},$$

since $1984 - j \equiv -j \pmod{992}$. The middle term $j = 992$, which is its own partner, also vanishes modulo 992. Hence $992 \mid S$. Because 1983 and 992 are coprime, $T \mid S$.

This method is algebra in a quotient ring together with a reflection symmetry of residue classes. The same strategy generalizes: instead of expanding a sum with thousands of terms, search for a transformation of the index that makes contributions cancel after passing to a suitable modulus.